

35/40 (6)

SIMATS ENGINEERING
ASSESSMENT TOOL 3

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COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0714

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2, BL6)

Assessment Tool Weightage: 30%

Total Marks: 25

QUESTIONS: 5

Annexure A

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows,

making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.

4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.

5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison-Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion

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DETAILED SOLUTIONS — COMPARATIVE ANALYSIS

Each of the five comparative-analysis statements below is expanded into a full solution: the comparison criteria are made explicit, a side-by-side comparison table is provided with supporting real-world examples/evidence, an analytical justification is given, and a concluding insight is drawn — addressing every rubric criterion (Understanding of Concepts, Comparison Depth, Criteria Selection, Analytical Skills, Organization, Use of Examples, Clarity, and Conclusion & Insights).

Question 1 — Star Topology vs. Bus Topology (Cost, Reliability, Performance)

1.1 Comparison Criteria Selected

Cost (hardware + cabling), Reliability (impact of a single failure), and Performance (behaviour as load/devices increase).

1.2 Side-by-Side Comparison

Criterion	Star Topology	Bus Topology
Hardware cost	Requires a central switch/hub plus one cable run per node — higher upfront hardware and cabling cost	Single shared backbone cable, no central device — lowest hardware cost
Installation cost/effort	More cable to purchase and pull (home-run to every node)	Minimal cabling; fast, cheap to install for small networks
Reliability (single fault)	A cable or node failure is isolated to that one device only — rest of network keeps working	A break anywhere in the backbone (or a bad terminator) can bring down the entire segment
Performance under load	Central switch forwards frames on a per-port basis, minimizing collisions even as devices are added	All devices share one medium — collisions and congestion increase noticeably as devices/traffic grow
Typical example	Office LAN with a 24-port managed switch (see Assessment Tool 2, Solution 1)	Legacy 10BASE2 coax lab network, or a classroom on a tight budget (see Assessment Tool 2, Solution 3)

1.3 Analytical Justification

- Cost driver: star topology's cost comes from the physical requirement of a dedicated link per node plus an active central device; bus topology avoids both, which is why it is favoured only where budget is the dominant constraint.
- Reliability driver: the difference is structural — star topology distributes failure risk across many independent links, while bus topology concentrates all traffic on one shared cable, making that single cable a critical point of failure.
- Performance driver: a switch in a star topology can forward multiple frames simultaneously on different ports (full-duplex, dedicated bandwidth per link), whereas bus topology forces every device to contend for the same shared channel using CSMA/CD-style access, so throughput per device falls as more devices are added.

1.4 Conclusion / Insight

Star topology trades higher upfront cost for significantly better reliability and performance at scale, making it the preferred choice for any network expected to grow beyond a handful of devices or that cannot tolerate a full network outage. Bus topology remains justifiable only for very small, low-budget, low-traffic, non-critical networks.

Question 2 — Mesh Topology vs. Star Topology (Fault Tolerance)

2.1 Comparison Criteria Selected

Fault tolerance is assessed along two dimensions: (a) resilience to a single link failure, and (b) resilience to failure of the most critical device in the topology.

2.2 Side-by-Side Comparison

Criterion	Mesh Topology	Star Topology
Path redundancy	Every node has multiple direct/indirect paths to every other node	Every node has exactly one path — through the central switch/hub
Single link failure	Traffic reroutes automatically through an alternate path; no disruption	If the failed link is a node's own cable, only that node is affected
Critical single point of failure	None — no single device or link is indispensable in a full mesh	The central switch/hub is a single point of failure — its failure stops the entire network
Typical fault-tolerance level	Very high — used where near-zero downtime is mandatory	Moderate — good for isolating node-level faults, but not hub-level faults
Typical example	Bank core network with redundant routers/switches (see Assessment Tool 2, Solution 2)	Startup office LAN with one managed switch (see Assessment Tool 2, Solution 1)

2.3 Analytical Justification

- Mesh topology's fault tolerance comes from having many redundant physical paths; routing protocols such as OSPF or BGP detect a failed link and recompute an alternate path within milliseconds, so the network as a whole never depends on any one link or device.
- Star topology only isolates faults that occur on the spokes (the individual node-to-switch cables); the hub itself is not redundant, so its failure is catastrophic for the whole network unless a secondary switch and dual-homing are added — which effectively begins turning the design into a hybrid/partial-mesh at the core.

2.4 Conclusion / Insight

Mesh topology is the superior choice specifically for mission-critical environments (banking, healthcare, emergency services) where the cost of downtime far outweighs the higher cabling/hardware cost. Star topology remains adequate wherever occasional, brief central-device downtime is an acceptable risk in exchange for a much simpler and cheaper design.

Question 3 — Bus Topology vs. Mesh Topology (Installation Complexity)

3.1 Comparison Criteria Selected

Installation complexity is compared using: number of physical connections required, cabling effort, and how complexity scales as devices (n) are added.

3.2 Side-by-Side Comparison

Criterion	Bus Topology	Mesh Topology
Connections required for n devices	n taps onto a single shared backbone (plus 2 terminators)	$n(n-1)/2$ direct links for a full mesh — grows quadratically
Cabling effort	One backbone run; each device taps in with a single connector	Every node needs a dedicated cable and port to every other node
Ports/interfaces needed per device	One (T-connector or single switch port)	$n-1$ (one interface per other node, in a full mesh)
Scaling behaviour	Adding a device is a quick tap onto the existing backbone	Adding one device requires n new links to every existing device
Typical example	16-seat classroom lab (see Assessment Tool 2, Solution 3)	Bank core layer with 4-6 core devices in partial mesh (see Assessment Tool 2, Solution 2)

3.3 Analytical Justification

- Bus topology's installation effort grows linearly with the number of devices (each new device is one more tap on the same backbone), which is why it remains simple even as a room grows modestly in size.
- Mesh topology's installation effort grows quadratically ($n(n-1)/2$ links for full mesh), so while it is very manageable for a handful of core devices (e.g., 4-6 devices in a bank's core layer), it becomes impractical for tens or hundreds of end devices — which is exactly why mesh is used only at the core/backbone layer and never at the access layer for large numbers of end-user devices.

3.4 Conclusion / Insight

Bus topology's simplicity makes it attractive only for small, low-cost networks, while mesh topology's installation complexity is the direct trade-off organizations accept in exchange for its superior fault tolerance — and in practice this cost is controlled by limiting full/partial mesh to a small number of critical core devices rather than the entire network.

Question 4 — Hybrid Topology vs. Star Topology (Scalability)

4.1 Comparison Criteria Selected

Scalability is compared on: how new segments/devices are added, whether the existing network is disrupted during expansion, and the ultimate ceiling on growth.

4.2 Side-by-Side Comparison

Criterion	Hybrid Topology	Star Topology
Basis for expansion	Combines multiple topologies (star, mesh, bus) so new segments can use whichever sub-topology fits	Limited to adding more devices to the same central switch/hub
Growth ceiling	Practically unlimited — new distribution/access layers or buildings can be added indefinitely	Bounded by the port count of the central device
Disruption during expansion	New segments (e.g., a new building's distribution switch) can be added without touching the existing structure	Once ports are exhausted, new hardware (a second switch) and re-cabling/uplinking is required
Typical example	University campus: core ring + per-building star + per-lab bus/star (see Assessment Tool 2, Solution 4)	20-person startup with one 24-port switch (see Assessment Tool 2, Solution 1)

4.3 Analytical Justification

- Hybrid topology's scalability comes from its hierarchical structure — a core/distribution/access model lets an organization add a new distribution switch and uplink it to the core whenever a new building or department is needed, without modifying anything already deployed.
- Star topology's scalability is fundamentally capped by the physical port count of its one central device; once that device is full, growth requires either replacing it with a larger switch or cascading a second switch — at which point the network has, in effect, begun evolving toward a hybrid design.

4.4 Conclusion / Insight

For organizations that expect sustained or unpredictable growth (a university campus, a multi-site company), hybrid topology is the more future-proof choice; a pure star topology remains excellent for a single, size-bounded location such as a small office, but scalability beyond that scope naturally pushes the design toward a hybrid architecture.

Question 5 — Maintenance Complexity Across Star, Mesh, Bus, and Hybrid Topologies

5.1 Comparison Criteria Selected

Maintenance is compared using: ease of fault identification, ease of fault isolation, and the level of administrative skill required for ongoing management.

5.2 Side-by-Side Comparison

Topology	Fault Identification	Fault Isolation	Admin Skill Required	Overall Maintenance Level
Star	Easy — switch port/link lights and status pages point directly to the affected node	Excellent — one bad cable/node does not affect others	Low-to-moderate	Easy
Bus	Hard — a break anywhere along the backbone must be traced with specialised tools (e.g., TDR)	Poor — the entire segment goes down and must be tested section by section	Moderate	Difficult
Mesh	Moderate — the fault itself is often masked by automatic rerouting, but many redundant links make locating the actual failed link time-consuming	Excellent for continuity, but tracing exactly which of many links failed is complex	High (routing protocols, redundant hardware)	Difficult
Hybrid	Varies by segment — star segments are easy, mesh/bus segments inherit their own difficulty	Good — failures are generally contained within the sub-topology/segment where they occur	High (must understand every combined topology in use)	Moderate

5.3 Analytical Justification

- Star topology is easiest to maintain because its point-to-point wiring maps directly onto switch port diagnostics — a technician can identify and isolate a fault to a single cable or node almost immediately.
- Bus topology is harder to maintain than star precisely because there is no central diagnostic point; a break must be physically traced along the backbone, and the entire segment is affected while the fault is located.
- Mesh topology is reliable in operation but difficult to maintain because its redundancy, which is its greatest strength, is also what makes isolating one specific failed link laborious among many overlapping paths.
- Hybrid topology's maintenance burden is moderate and segment-dependent: administrators benefit from the easy-to-maintain star segments but must also retain the specialised skills needed for whichever mesh or bus segments are combined into the design.

5.4 Conclusion / Insight

Overall maintenance complexity ranks, from easiest to hardest: Star < Hybrid < Bus ≈ Mesh. This means the topology that is simplest to operate day-to-day (star) is not always the one that is most

reliable under failure (mesh); network designers must balance ease of maintenance against fault tolerance requirements when choosing a topology, which is exactly why large organizations adopt hybrid designs — accepting moderate maintenance complexity in exchange for combining the strengths of multiple topologies where each is most appropriate.